

- Q.34 Write a short note on nebulizers and humidifiers.
Q.35 Explain Ophthalmoscope withy diagram.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain Multi-Parameter Monitor with diagram.
Q.37 Explain blood pressure measurement methods with diagram.
Q.38 Write note on following:
a) Hearing Aid b) Respiratory volumes

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Roll No.

3rd Sem / Med Eltx.

Sub : Biomedical Instrumentation-I /

Biomed. Instr./ Basic of Instr.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A sphygmomanometer is used to measure.
a) Blood Glucose b) Blood Pressure
c) Blood Oxygen d) Blood volume
- Q.2 Which electrode is commonly used for ECG recordings.
a) Needle electrode b) Surface electrode
c) Suction electrode d) All of above
- Q.3 Recording of Electrical activity of Brain is called
a) EFG b) EEG
c) EMG d) EOG
- Q.4 ECG Signal is usually measured in
a) Volts b) Millivolts
c) Microvolts d) Ampere
- Q.5 The purpose of defibrillator is
a) Monitor blood flow
b) Detect Arrhythmia
c) Restart a heart beat
d) Measure pulse

- Q.6 Which of the following is not a biomedical sensor
- Photoplethysmograph
 - Thermistor
 - Tachometer
 - Piezoelectric sensor
- Q.7 Bio Electric signal generated by the retina
- ECG
 - EEG
 - EMG
 - ERG
- Q.8 The Piezoelectric is primarily used in which biomedical instrument
- ECG
 - Patient Monitor
 - Ultrasound machine
 - Defibrillator
- Q.9 A bipolar lead system in ECG uses is
- One electrode
 - Two electrode with one ground
 - Two active electrode
 - None
- Q.10 The typical energy range used by a biphasic defibrillator is
- 5-10 J
 - 10-50 J
 - 120-200 J
 - 300-400 J

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Bio-Electric signal
- Q.12 Transducer.
- Q.13 Depolarization.

- Q.14 Expand VCG.
- Q.15 Heart Rate
- Q.16 Brain waves.
- Q.17 Neuromuscular junction
- Q.18 What is called Natural Pacemaker of body?
- Q.19 Cardiac output.
- Q.20 Motion Artefact.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a note on role of biomedical engineer in healthcare.
- Q.22 Explain a basic medical instrumentation system with diagram.
- Q.23 Explain cardiac system with diagram.
- Q.24 Explain ECG, Lead system with diagram.
- Q.25 Explain effect of artifact in ECG measurements.
- Q.26 Write a note on Holter monitor.
- Q.27 Explain Central nervous system with diagram.
- Q.28 Write a short note on blood sugar measurement.
- Q.29 Explain 10-20 Electrode placement system with diagram.
- Q.30 Write a note on Neuromuscular system.
- Q.31 Explain infusion pump with diagram.
- Q.32 Explain ambulatory monitoring system.
- Q.33 Explain various electrodes used in EMG measurements.